# Arquitetura de Computadores

## Aula 1 – Fundamentos de Arquitetura de Computadores (Bit, Byte, Hexa)

DO QUE OS COMPUTADORES
SE ALIMENTAM?

SISTEMA DE NUMERAÇÃO

# $\text{Bases}_{10}$

- Decimal: 0 1 2 3 4 5 6 7 8 9
- 1337

M C D U

1337

- U → UNIDADE
- D → DEZENA
- C → CENTENA
- M → MILHAR

$$
\begin{array}{r}
1000 \\
300 \\
+\quad 30 \\
7 \\
\hline
1337
\end{array}
$$

$$1000 \rightarrow 1 \times 10^3 = 1 \times 1000 = 1000$$
$$300 \rightarrow 3 \times 10^2 = 3 \times 100 = 300$$
$$30 \rightarrow 3 \times 10^1 = 3 \times 10 = 30$$
$$7 \rightarrow 7 \times \underbrace{10}_{\text{BASE}}{}^0 = 7 \times 1 = 7$$

$$
\begin{array}{r}
+\quad 7 \\
\hline
1337
\end{array}
$$

M C D U

$$3214 = 4 + 10 + 200 + 3000 = 3214$$

# Bases$_2$

- Binária: 0 1
- BIT ← MENOR UNIDADE
- BYTE ← 8bits → 1 BYTE
- Bit + significativo
- Bit - significativo
- 10100111001

ENDIANESS → BIG ENDIAN + 43 −

→ LITTLE ENDIAN − 34 +

+ 1337 −

11bits

8bits 1 BYTE

1337 (BE)

7331 (LE)

# Bases

- Binária: 0 1
- BIT
- BYTE
- Bit + significativo
- Bit - significativo
- $10100111001_2$

→ DECIMAL

$$1 \times 2^{10} = 1 \times 1024 = 1024$$
$$\cancel{0 \times 2^{9} = 0 \times 512}$$
$$1 \times 2^{8} = 1 \times 256 = 256$$
$$\cancel{0 \times 2^{7} = 0 \times 128}$$
$$\cancel{0 \times 2^{6} = 0 \times 64}$$
$$1 \times 2^{5} = 1 \times 32 = 32$$
$$1 \times 2^{4} = 1 \times 16 = 16$$
$$1 \times 2^{3} = 1 \times 8 = 8$$
$$\cancel{0 \times 2^{2} = 0 \times 4}$$
$$\cancel{0 \times 2^{1} = 0 \times 2}$$
$$1 \times 2^{0} = 1 \times 1 = 1$$
$$+ \; 1337_{10}$$

$2^0 = 1$
$2^1 = 2$
$2^2 = 4$
$2^3 = 8$
$2^4 = 16$
$2^5 = 32$
$2^6 = 64$
$2^7 = 128$
$2^8 = 256$
$2^9 = 512$
$2^{10} = 1024$

| 1024 | ~~512~~ | 256 | ~~128~~ | ~~64~~ | 32 | 16 | 8 | ~~4~~ | ~~2~~ | 1 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | ~~0~~ | 1 | ~~0~~ | ~~0~~ | 1 | 1 | 1 | ~~0~~ | ~~0~~ | 1 | = 1337 |

# Bases$_8$

- Octal: 0 1 2 3 4 5 6 7 10 11 12 13 14 15 16 17 20
- 2471$_8$ → decimal

$1 \times 8^1 = 1 \times 8 = 8$

$0 \times 8^0 = 0 \times 1 = 0$ +

$8_{10}$

$2 \times 8^3 = 2 \times 512 = 1024$

$4 \times 8^2 = 4 \times 64 = 256$

$7 \times 8^1 = 7 \times 8 = 56$

$1 \times 8^0 = 1 \times 1 = 1$

+ $1337_{10}$

$8^0 = 1$

$8^1 = 8$

$8^2 = 64$

$8^3 = 512$

$$\begin{array}{cccc} 512 & 64 & 8 & 1 \\ \times 2 & \times 4 & \times 7 & \times 1 \\ \hline 1024 & +\ 256 & +\ 56 & +\ 1 \end{array} = 1337_{10}$$

# Bases 16/H

- Hexadecimal: 0 1 2 3 4 5 6 7 8 9 A B C D E F 10 … 1F 20

  10 11 12 13 14 15

- $539_H$ → DECIMAL

$$5 \times 16^2 = 5 \times 256 = 1280$$
$$3 \times 16^1 = 3 \times 16 = 48$$
$$9 \times 16^0 = 9 \times 1 = 9$$
$$+\ 1337_{10}$$

$16^0 = 1$
$16^1 = 16$
$16^2 = 256$
$16^3 = 4096$
$16^4 = 65536$

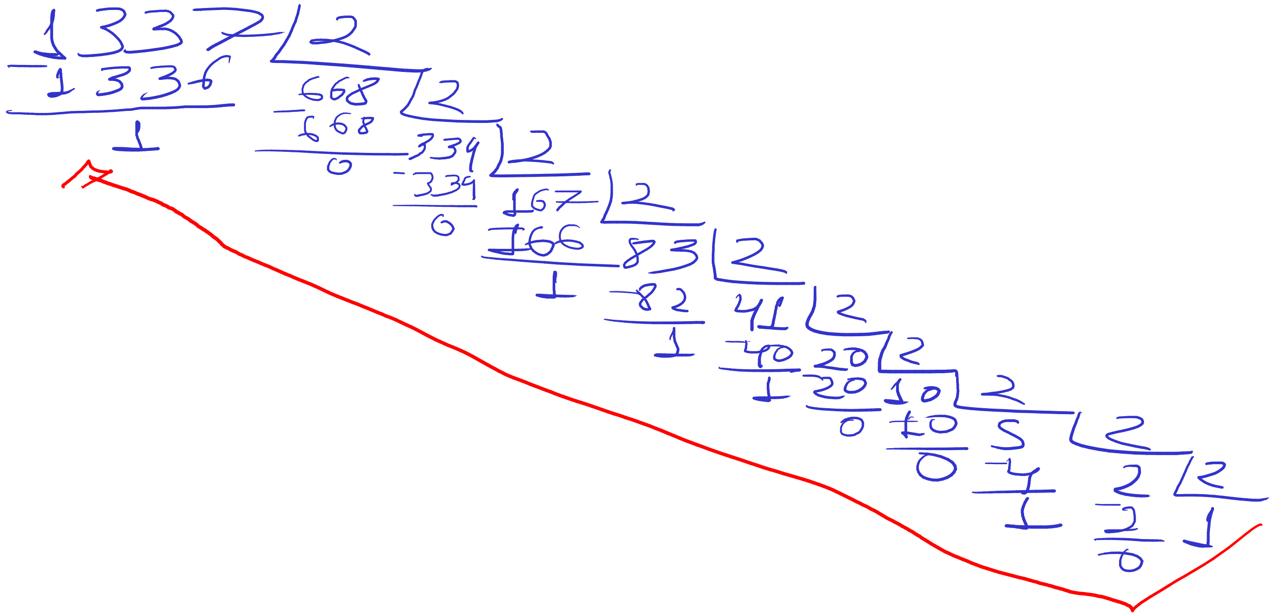
B∅CA

$$\frac{\overset{256}{\times 5}}{1280} + \frac{\overset{16}{\times 3}}{48} + \frac{\overset{1}{\times 9}}{9} = 1337_{10}$$

$$\frac{\overset{4096}{\times B_{11}}}{45056} + \frac{\overset{256}{\times \emptyset}}{0} + \frac{\overset{16}{\times C_{12}}}{192} + \frac{\overset{1}{\times A_{10}}}{10} = 45258_{10}$$

# Conversões de base

- Binário ↔ Decimal




1 0 1 0 0 1 1 1 0 0 1

# Conversões de base

- Binário ↔ Octal

$6 \div 2 = 3$, resto $0$; $3 \div 2 = 1$, resto $1$

1 1 0

| 4 | 2 | 1 | |
|---|---|---|---|
| 1 | 1 | 0 | → 6 |

| 8 | 4 | 2 | 1 | |
|---|---|---|---|---|
| 1 | 1 | 1 | 0 | → 14 |

| DEC | OCT | HEX | BIN (8 4 2 1) |
|---|---|---|---|
| 0 | 0 | 0 | 000 |
| 1 | 1 | 1 | 001 |
| 2 | 2 | 2 | 010 |
| 3 | 3 | 3 | 011 |
| 4 | 4 | 4 | 100 |
| 5 | 5 | 5 | 101 |
| 6 | 6 | 6 | 110 |
| 7 | 7 | 7 | 111 |
| 8 | | 8 | 1000 |
| 9 | | 9 | 1001 |
| 10 | | A | 1010 |
| 11 | | B | 1011 |
| 12 | | C | 1100 |
| 13 | | D | 1101 |
| 14 | | E | 1110 |
| 15 | | F | 1111 |

$2^0 = 1$
$2^1 = 2$
$2^2 = 4$
$2^3 = 8$
$2^4 = 16$
$2^5 = 32$
$2^6 = 64$
$2^7 = 128$
$2^8 = 256$
$2^9 = 512$
$2^{10} = 1024$

$8^0 = 1$
$8^1 = 8$
$8^2 = 64$
$8^3 = 512$

$16^0 = 1$
$16^1 = 16$
$16^2 = 256$
$16^3 = 4096$
$16^4 = 65536$

# Conversões de base

- Binário ↔ Octal

0 1 0 | 1 0 0 | 1 1 1 | 0 0 1

$2 \quad 4 \quad 7 \quad 1_8$

| Dec | Oct | Hex | Bin (8 4 2 1) |
|---|---|---|---|
| 0 | 0 | 0 | 000 |
| 1 | 1 | 1 | 001 |
| 2 | 2 | 2 | 010 |
| 3 | 3 | 3 | 011 |
| 4 | 4 | 4 | 100 |
| 5 | 5 | 5 | 101 |
| 6 | 6 | 6 | 110 |
| 7 | 7 | 7 | 111 |
| 8 | | 8 | 1000 |
| 9 | | 9 | 1001 |
| 10 | | A | 1010 |
| 11 | | B | 1011 |
| 12 | | C | 1100 |
| 13 | | D | 1101 |
| 14 | | E | 1110 |
| 15 | | F | 1111 |

# Conversões de base

- Binário ↔ Hexadecimal

0101 | 0011 | 1001

5 3 9 H

1F → 1 BYTE

22 → 1 BYTE

00 → 8 bits → 1 BYTE

BE → DE AD BE EF

LE → EF BE AD DE

| DEC | OCT | HEX | BIN (8 4 2 1) |
|---|---|---|---|
| 0 | 0 | 0 | 0000 |
| 1 | 1 | 1 | 0001 |
| 2 | 2 | 2 | 0010 |
| 3 | 3 | 3 | 0011 |
| 4 | 4 | 4 | 0100 |
| 5 | 5 | 5 | 0101 |
| 6 | 6 | 6 | 0110 |
| 7 | 7 | 7 | 0111 |
| 8 | | 8 | 1000 |
| 9 | | 9 | 1001 |
| 10 | | A | 1010 |
| 11 | | B | 1011 |
| 12 | | C | 1100 |
| 13 | | D | 1101 |
| 14 | | E | 1110 |
| 15 | | F | 1111 |

R 1BYTE G 1BYTE B 1BYTE

BE → FØ FØ CA = 3BYTES
24bits

LE → CA FØ FØ

# ASCII TABLE

| Decimal | Hexadecimal | Binary | Octal | Char | Decimal | Hexadecimal | Binary | Octal | Char | Decimal | Hexadecimal | Binary | Octal | Char |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | [NULL] | 48 | 30 | 110000 | 60 | 0 | 96 | 60 | 1100000 | 140 | ` |
| 1 | 1 | 1 | 1 | [START OF HEADING] | 49 | 31 | 110001 | 61 | 1 | 97 | 61 | 1100001 | 141 | a |
| 2 | 2 | 10 | 2 | [START OF TEXT] | 50 | 32 | 110010 | 62 | 2 | 98 | 62 | 1100010 | 142 | b |
| 3 | 3 | 11 | 3 | [END OF TEXT] | 51 | 33 | 110011 | 63 | 3 | 99 | 63 | 1100011 | 143 | c |
| 4 | 4 | 100 | 4 | [END OF TRANSMISSION] | 52 | 34 | 110100 | 64 | 4 | 100 | 64 | 1100100 | 144 | d |
| 5 | 5 | 101 | 5 | [ENQUIRY] | 53 | 35 | 110101 | 65 | 5 | 101 | 65 | 1100101 | 145 | e |
| 6 | 6 | 110 | 6 | [ACKNOWLEDGE] | 54 | 36 | 110110 | 66 | 6 | 102 | 66 | 1100110 | 146 | f |
| 7 | 7 | 111 | 7 | [BELL] | 55 | 37 | 110111 | 67 | 7 | 103 | 67 | 1100111 | 147 | g |
| 8 | 8 | 1000 | 10 | [BACKSPACE] | 56 | 38 | 111000 | 70 | 8 | 104 | 68 | 1101000 | 150 | h |
| 9 | 9 | 1001 | 11 | [HORIZONTAL TAB] | 57 | 39 | 111001 | 71 | 9 | 105 | 69 | 1101001 | 151 | i |
| 10 | A | 1010 | 12 | [LINE FEED] | 58 | 3A | 111010 | 72 | : | 106 | 6A | 1101010 | 152 | j |
| 11 | B | 1011 | 13 | [VERTICAL TAB] | 59 | 3B | 111011 | 73 | ; | 107 | 6B | 1101011 | 153 | k |
| 12 | C | 1100 | 14 | [FORM FEED] | 60 | 3C | 111100 | 74 | < | 108 | 6C | 1101100 | 154 | l |
| 13 | D | 1101 | 15 | [CARRIAGE RETURN] | 61 | 3D | 111101 | 75 | = | 109 | 6D | 1101101 | 155 | m |
| 14 | E | 1110 | 16 | [SHIFT OUT] | 62 | 3E | 111110 | 76 | > | 110 | 6E | 1101110 | 156 | n |
| 15 | F | 1111 | 17 | [SHIFT IN] | 63 | 3F | 111111 | 77 | ? | 111 | 6F | 1101111 | 157 | o |
| 16 | 10 | 10000 | 20 | [DATA LINK ESCAPE] | 64 | 40 | 1000000 | 100 | @ | 112 | 70 | 1110000 | 160 | p |
| 17 | 11 | 10001 | 21 | [DEVICE CONTROL 1] | 65 | 41 | 1000001 | 101 | A | 113 | 71 | 1110001 | 161 | q |
| 18 | 12 | 10010 | 22 | [DEVICE CONTROL 2] | 66 | 42 | 1000010 | 102 | B | 114 | 72 | 1110010 | 162 | r |
| 19 | 13 | 10011 | 23 | [DEVICE CONTROL 3] | 67 | 43 | 1000011 | 103 | C | 115 | 73 | 1110011 | 163 | s |
| 20 | 14 | 10100 | 24 | [DEVICE CONTROL 4] | 68 | 44 | 1000100 | 104 | D | 116 | 74 | 1110100 | 164 | t |
| 21 | 15 | 10101 | 25 | [NEGATIVE ACKNOWLEDGE] | 69 | 45 | 1000101 | 105 | E | 117 | 75 | 1110101 | 165 | u |
| 22 | 16 | 10110 | 26 | [SYNCHRONOUS IDLE] | 70 | 46 | 1000110 | 106 | F | 118 | 76 | 1110110 | 166 | v |
| 23 | 17 | 10111 | 27 | [END OF TRANS. BLOCK] | 71 | 47 | 1000111 | 107 | G | 119 | 77 | 1110111 | 167 | w |
| 24 | 18 | 11000 | 30 | [CANCEL] | 72 | 48 | 1001000 | 110 | H | 120 | 78 | 1111000 | 170 | x |
| 25 | 19 | 11001 | 31 | [END OF MEDIUM] | 73 | 49 | 1001001 | 111 | I | 121 | 79 | 1111001 | 171 | y |
| 26 | 1A | 11010 | 32 | [SUBSTITUTE] | 74 | 4A | 1001010 | 112 | J | 122 | 7A | 1111010 | 172 | z |
| 27 | 1B | 11011 | 33 | [ESCAPE] | 75 | 4B | 1001011 | 113 | K | 123 | 7B | 1111011 | 173 | { |
| 28 | 1C | 11100 | 34 | [FILE SEPARATOR] | 76 | 4C | 1001100 | 114 | L | 124 | 7C | 1111100 | 174 | \| |
| 29 | 1D | 11101 | 35 | [GROUP SEPARATOR] | 77 | 4D | 1001101 | 115 | M | 125 | 7D | 1111101 | 175 | } |
| 30 | 1E | 11110 | 36 | [RECORD SEPARATOR] | 78 | 4E | 1001110 | 116 | N | 126 | 7E | 1111110 | 176 | ~ |
| 31 | 1F | 11111 | 37 | [UNIT SEPARATOR] | 79 | 4F | 1001111 | 117 | O | 127 | 7F | 1111111 | 177 | [DEL] |
| 32 | 20 | 100000 | 40 | [SPACE] | 80 | 50 | 1010000 | 120 | P | | | | | |
| 33 | 21 | 100001 | 41 | ! | 81 | 51 | 1010001 | 121 | Q | | | | | |
| 34 | 22 | 100010 | 42 | " | 82 | 52 | 1010010 | 122 | R | | | | | |
| 35 | 23 | 100011 | 43 | # | 83 | 53 | 1010011 | 123 | S | | | | | |
| 36 | 24 | 100100 | 44 | $ | 84 | 54 | 1010100 | 124 | T | | | | | |
| 37 | 25 | 100101 | 45 | % | 85 | 55 | 1010101 | 125 | U | | | | | |
| 38 | 26 | 100110 | 46 | & | 86 | 56 | 1010110 | 126 | V | | | | | |
| 39 | 27 | 100111 | 47 | ' | 87 | 57 | 1010111 | 127 | W | | | | | |
| 40 | 28 | 101000 | 50 | ( | 88 | 58 | 1011000 | 130 | X | | | | | |
| 41 | 29 | 101001 | 51 | ) | 89 | 59 | 1011001 | 131 | Y | | | | | |
| 42 | 2A | 101010 | 52 | * | 90 | 5A | 1011010 | 132 | Z | | | | | |
| 43 | 2B | 101011 | 53 | + | 91 | 5B | 1011011 | 133 | [ | | | | | |
| 44 | 2C | 101100 | 54 | , | 92 | 5C | 1011100 | 134 | \ | | | | | |
| 45 | 2D | 101101 | 55 | - | 93 | 5D | 1011101 | 135 | ] | | | | | |
| 46 | 2E | 101110 | 56 | . | 94 | 5E | 1011110 | 136 | ^ | | | | | |
| 47 | 2F | 101111 | 57 | / | 95 | 5F | 1011111 | 137 | _ | | | | | |